

called *quantitative RT-PCR* (*qRT-PCR*), can detect the light and measure the PCR product, thus avoiding the need for electrophoresis while providing quantitative data, a distinct advantage. RT-PCR or qRT-PCR can also be carried out with mRNAs collected from different tissues at one time to discover which tissue is producing a specific mRNA.

➔ **Instructors:** The Scientific Skills Exercise “Analyzing Quantitative and Spatial Gene Expression Data” can be assigned in **Mastering Biology**. Students can work with data from an experiment that investigated mRNA expression using both *in situ* hybridization and quantitative RT-PCR.

Studying the Expression of Interacting Groups of Genes

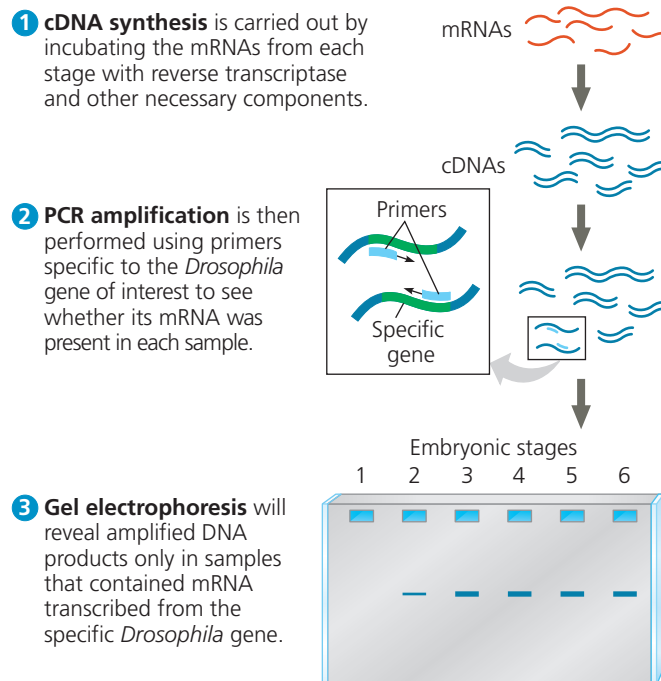
A major goal of biologists is to learn how genes act together to produce and maintain a functioning organism. Now that

▼ Figure 20.11 Research Method

RT-PCR Analysis of the Expression of Single Genes

Application RT-PCR uses the enzyme reverse transcriptase (RT) in combination with PCR and gel electrophoresis. RT-PCR can be used to compare gene expression between samples—for instance, in different embryonic stages, in different tissues, or in the same type of cell under different conditions.

Technique In this example, samples containing mRNAs from six embryonic stages of *Drosophila* were analyzed for a specific mRNA as shown below. (In steps 1 and 2, the mRNA from only one stage is shown.)



the genomes of a number of species have been sequenced, it is possible to study the expression of large groups of genes—an approach called the *systems approach*. Researchers use what is known about the whole genome to investigate which genes are transcribed in different tissues or at different stages of development. One aim is to identify networks of gene expression across an entire genome.

To accomplish such genome-wide expression studies, today’s rapid, inexpensive DNA-sequencing methods allow researchers to discover which genes are expressed by simply sequencing the cDNA samples from different tissues or different embryonic stages. This straightforward method is called **RNA sequencing**, or **RNA-seq** (pronounced “RNA-seek”), even though it is the cDNA that is actually sequenced. In RNA-seq, the mRNA (or other RNA) samples are isolated, cut into shorter, similar-sized fragments, and converted into cDNAs (**Figure 20.12**). These short cDNA stretches are sequenced, and a computer program reassembles them, either mapping them onto the genome of the species in question (when available) or simply putting the fragments in order from scratch based on overlapping sequences of multiple RNAs.

▼ **Figure 20.12 Use of RNA sequencing (RNA-seq) to analyze expression of many genes.** RNA-seq yields a wide range of information about expression of genes, including their level of expression.

